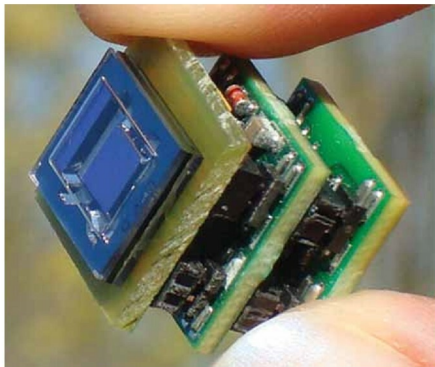


WORKING WITH TEMPERATURE SENSORS: A GUIDE

Real-time temperature monitoring with dedicated temperature sensors ensure that today's smaller and faster systems operate in the safe thermal zone. The new-generation sensors monitor internal and external components' hot spots with pinpoint accuracy. Availability of accurate, low-cost and easy-to-use sensor ICs permits designers to make on-chip temperature measurements to squeeze the maximum performance from their systems



Fully autonomous wireless temperature sensor powered by a vibrational energy harvester

■ T.K. HAREENDRAN

Temperature is the most often measured environmental quantity and many biological, chemical, physical, mechanical and electronic systems are affected by temperature. Some processes work well only within a narrow range of temperatures. So proper care must be taken to monitor and protect the system.

When temperature limits are exceeded, electronic components and circuits may be damaged by exposure to high temperatures. Temperature sensing helps to enhance circuit stability. By sensing the temperature inside the equipment, high temperature levels can be detected and actions can be taken to reduce system temperature, or even shut the system down to avert disasters.

Several temperature sensing techniques are used currently. The most common of these are thermocouples, thermistors and sensor integrated circuits (ICs). What is most suitable for your application depends on the required temperature range, linearity, accuracy, cost, features and the ease of designing the necessary support circuitry.

Thermocouples

A thermocouple consists of two dissimilar metals joined together at one end, to produce a small unique voltage at a given temperature. The thermoelectric voltage, resulting from the temperature difference from one end of the wire to the other, is actually the sum of all the voltage differences along the wire from end to end.

Thermocouples are available in different combinations of metals or calibrations. The four most common calibrations are J, K, T and E. Each calibration has a different temperature range and environment, although the maximum temperature varies with the diameter of the wire used in the thermocouple. For example, a 'type J' thermocouple is made from iron and



K-type thermocouple